The fluorophore absorbs photons leading to electrons moving to a higher energy state (excited state). When the electrons return to the ground state by losing energy, the fluorophore emits light of a longer wavelength (Figure 2.5). Three of the most common fluorophores used are Diamidino – phenylindole (DAPI) (emits blue), FITC (emits green), and Texas Red (emits red).

2.2.1 Principle

Light source such as Xenon or Mercury Arch Lamp which provides light in a wide range of wavelength, from ultraviolet to the infrared is directed through on exciter filter (selects the excitation wavelength). This light is reflected toward the sample by a special mirror called a dichroic mirror, which is designed to reflect light only at the excitation wavelength. The reflected light passes through the objective where it is focused onto the fluorescent specimen. The emissions from the specimen are in turn, passed back up through the objective where magnification of the image occurs and through the dichroic mirror.

This light is filtered by the barrier filter, which selects for the emission wavelength and filters out contaminating light from the arc lamp or other sources that are reflected off from the microscope components. Finally, the filtered fluorescent emission is sent to a detector where the image can be digitized.

2.2.2 Components of Fluorescence Microscope

The main components of the fluorescent microscope resemble the traditional light microscope. However, the two main difference are the type of light source used and the use of the specialized filter elements (Figure 2.6).

Infobits

Some substances have the property of luminescence. They emit light of one colour when exposed to light of a different colour. If light emission occurs within one millionth of a second of light exposure, the luminescence is phosphorescence. The colour of the emitted light has a longer wavelength than the colour of the exciting light. For example, fluorescein isothiocyanate (FITC) is excited by blue light and emits green light; rhodamine isothiocyanate is excited by green light and emits red light.

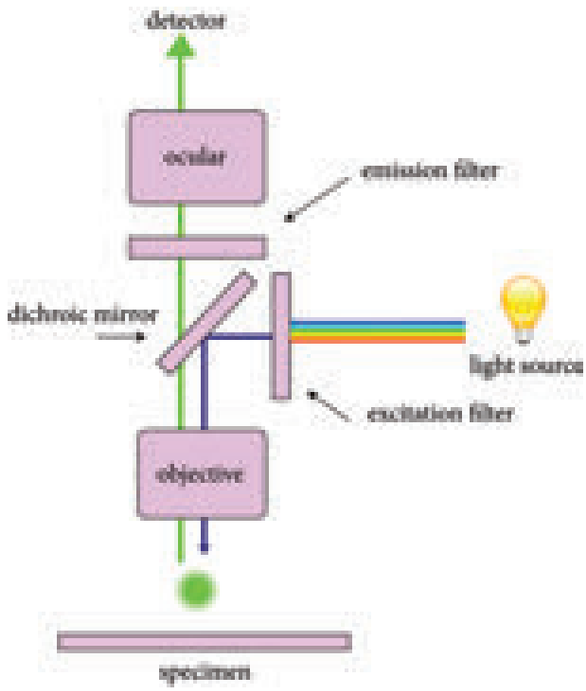


Figure 2.6: Components of fluorescence microscope

TN_GOVT_XII_Micro_Biology_CH02.indd 17 27-02-2019 13:44:12